Market revenue Declining 1-2% per year In 10 yrs to 2003, total industry profit pool fell 50%

Costs reduction strategy 2-3% per year every year

SKUs 6,000 → **14,200** → 5,000 → 9,000

Variety-> complexity, cost
SKUs within the product range had escalated from 6,000 (1997) to over 14,200 (2004)

1993
So popular, new products were delayed into production to limit growth to 8-10%

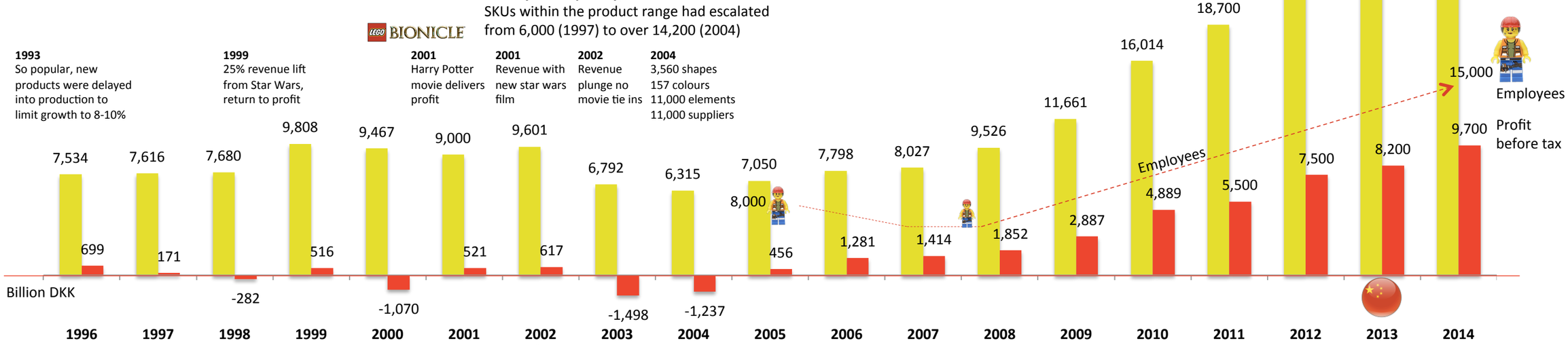
1999
25% revenue lift from Star Wars, return to profit

2001
Harry Potter movie delivers profit

2001
Revenue with new star wars film

2002
Revenue plunge no movie tie ins

2004
3,560 shapes
157 colours
11,000 elements
11,000 suppliers



1990s, lost edge in manufacturing and supply chain management



Green added as the 6th color in 1980s (took 15 yrs to achieve)

First loss since 1932



Strategy response
- Reduce costs 1b DK
- Reduce 10% jobs (1,000)
- Star Wars licencing massive revenue lift



No cost controls

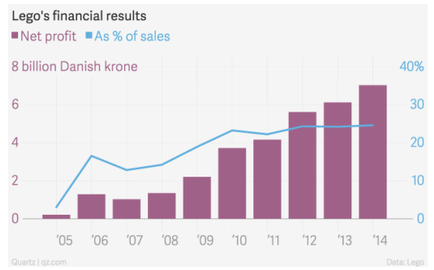
Outsourcing to an Electronics Manufacturing Services company, Flextronics, for 3 years

- Close US based factory in Enfield, Connecticut and relocate to Flextronics' in Mexico. (300 jobs)
- Distribution relocate to Mexico
- Billund retains technically more demanding products (LEGO Technic and BIONICLE).
- LEGO System products will relocate from Billund to Flextronics' facilities in Eastern Europe. (90/1,200 jobs gone)
- Flextronics takes over operations at the LEGO Group's factory in Kladno, the Czech Republic.
- Outsource European distribution

After in depth analysis of supply chain, strategy to achieve:
- headcount reduction 3,000 employees of 8,300
- products characterized by multiple labor-intensive decorating and assembly or packing processes will be outsourced because their high degree of uniformity and large production volumes render outsourcing profitable, partly because the packing of these products is very wage intensive.

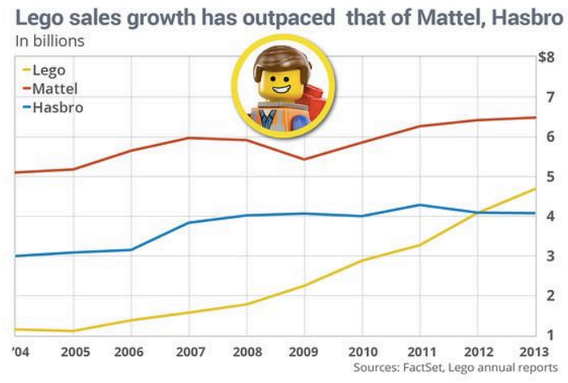
Discontinue outsourcing

Start to build new factory and distribution centre in China



Crisis because of:
- poor marketing (strategy) and
- poor supply chain management.

Solution:
- Supply chain management
- Better marketing (strategy)



Extension into new product areas (and exits)
- 1993: children's wear
- 1996: watches
- 1996: theme Park opened Windsor UK
- 1996: lego.com
- 1996: develop videogames
- 1996: LegoMedia in London – movies, TV, books

1998
First ever loss since 1932
- 1998: Mindstorms robotic bricks
- 1999: Legoland California
- 2000: cutback watches and publishing
- 2002: Duplo rebranded Lego Explore (unsuccesful)
- 2002: Legoland Germany

2004
New CEO
- 2005: sell theme parks. Keep 25% of mgt company
- 2005: shut videogames
- 2005: return Duplo Branding, reintro Lego City, improve mindstorms, discontinue Lego Clickits for girls
- 2006: Outsource manufacturing to Flextronics
- 2009: Take back manufacturing from Flextronics
- 2013: Build new factory and distribution centre in China

Operations Processes

In the last 10 years, Lego has grown into nothing less than the Apple of toys: a profit-generating, design-driven miracle built around premium, intuitive, highly covetable hardware that fans can't get enough of. Last year, fueled in part by The Lego Movie's Pixar-size popularity, the privately held company briefly surged ahead of rival Mattel to become the biggest toy manufacturer in the world, reporting first-half profits of $273 million on revenue of $2.03 billion

It's a remarkable achievement, particularly considering that Mattel makes a huge range of products—including -Barbie, Hot Wheels, Fisher-Price, and even the Lego knockoffs Mega Bloks—while Lego mostly sticks to variations on a single toy.

The company has recently moved beyond its core markets of Europe and North America with a major push into Asia, where it saw double-digit growth in 2013. A brand-new manufacturing and distribution center in China is being built solely to meet growing demand on that side of the planet. "

Ethical
Extra pieces or pieces from boxes that are caught on the line and identified as missing pieces or have boxes that are slightly damaged are used for donation boxes that are distributed to underprivileged children's organizations around the world.

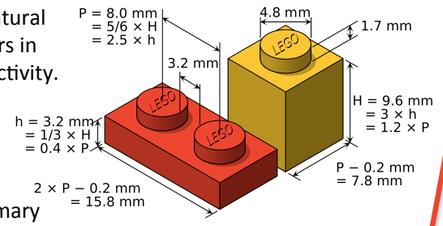
Transform**ED** resources

Regular checks on entire supply chain to ensure no deviations from established safety standards.

All suppliers were required to check the physical and chemical properties of the raw materials they shipped to LEGO, and certify that all the requisite tests had been carried out satisfactorily. Regular inspections of production are also carried out by LEGO's Corporate Quality Control Department

Transform**ING** processes

Lego bricks are manufactured through a molding process, where ABS plastic is heated and injected into standard molds and left to cool for about seven seconds. The molds are extremely accurate in that they only allow for a natural variation of 0.001 millimeters in each brick, to ensure connectivity.



Operations are 24/7

Leading edge technology an advantage compared to primary competitor, Canadian MegaBlocks now owned by Mattel.

Outputs

Billund:
- 30,000 bricks/minute,
- 21 billion bricks/yr
- 300 million tyres/yr (world's top tyre manufacturer)

Lego Universe: massive online game
- (biggest ever single product investment). Closed in 2012 after 2 yrs because unsatisfactory revenue

Lego Minifigs Online: massine online game released Dec 14

Recall goal: 0 per year
- Last recall 2009, and one in 2002
- One recall in the last decade

World's largest toy manufacturer.

All in all, Lego factories produce 33,000 bricks every minute, for a total of 16 billion bricks every year. The manufacturing facilities can make 3,000 different types of elements, including 300 million tires, and these pieces go into 37,000 Lego sets every hour.

Variety: large boxes have to be manually packed

Inputs

- plastic granules composed primarily of acrylonitrile butadiene styrene (ABS).
- When the ABS granules arrive at Lego manufacturing facilities, they're vacuumed into several storage silos
- The average Lego plant has about 14 silos, and each can hold 33 tons of ABS granules
- Billund use 60 tonne/day, and one week's supply is held on site (420 tonne)
- 11,000 suppliers (2004). Resin-sourcing analyzes helped to cut its resin costs in half and reduce supplier number by 80 percent.

Variation in demand
- Sell 50% of volume in 6 weeks before xmas. Must build inventories during year and release in Nov/Dec to retailers
- Can mould>decorate>package in 10 days to respond to demand

Flexibility
- Moulding machines can mould any piece with correct mould
- Decorating line can decorate any piece with correct software

Transformation process

- Granules in 14 silos,
- 12 production lines
- 23m high storage warehouse
- granules travel through tubes to the injection molding machines (made in Germany and large LEGO maintenance team every 3 weeks)
- Automated machines use very accurate molds. 2 hours to change moulds (150kg-1500kg)
- Granules melted at temperatures of up to 450 degrees F (232 degrees C), injected into molds and apply between 25 and 150 tons of pressure. After about seven seconds, the new Lego pieces cool and fall onto a conveyor.
- The injection-molding process uses large, heavy molds that are manufactured in Germany.
- Molded elements fall into bins and wait for a robot to carry them to the assembly hall.
- When the bin fills, the molding machine signals a robot to pick it up and carry it to an assembly hall.
- In the Billund factory, eight robots move 600 bins of elements per hour. In the assembly hall, machines stamp designs onto bricks and assemble components that require multiple pieces.
- 9 hour buffer on stock on factory floor.

Facilities

- Klando, Czech Republic
- Billund, Denmark (largest plastic moulding factory in world)
- Mexico
- China

Quality control

- **Defect rate 18 per million bricks** 0.00002% fail
- Quality assurance testers also perform numerous inspections and tests on Lego elements.
- Machines perform drop, torque, tension, compression, saliva, bite and impact tests to make sure the toys are sturdy and safe. Random blocks from line selected for testing in Quality Control Room. Packets are weight tested 8 times.
- Clutch power tested to ensure grip to other blocks.
- Technicians use a measuring beaker to determine whether pieces could cause a choking hazard for small children.
- Global quality control standards

Each LEGO set element of (from bricks to the instruction manual) through rigorous tests to make sure that they follow company, consumer, and international standards. Lego also looks for new ways to improve their production process from both a business and environmental perspective.

Logistics

The number of logistics service providers was cut from 26 to 4. (2004)

Sequencing and scheduling

Manufacturing: Lego rationalized production lines, reducing the number of products that could be made on each machine, adding simplicity and reducing changeover costs. It established more fixed production cycles, and better tied manufacturing to the rest of the supply chain. For example, it was no longer acceptable to make manual changes in a molding machine without informing the finished-goods packing team, an important consideration since different kits are packaged in different boxes.

New product development
130 new sets per year
Moulds 50-300k Euro, made in Germany

Typical product life cycle 3 years
Over past decade dramatically decrease the time it takes to develop and produce a new set.
Average development period now 12 months from start to finish, down from 2-3 years.